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Inventor(s)

FODOR, JR.; Gregory et al.

ADAPTIVE ORIENTATION CONTROL FOR INTERACTIVE ARTIFICIAL INTELLIGENCE SYSTEMS

Abstract

A system may include a sensor, a display, and a processor. The sensor may generate sensor data based on a factor of the system. The display may display an animated face of an artificial intelligence (AI) agent. The processor may be coupled to the sensor and the display. The processor may determine an orientation of the system based on the sensor data. The processor may also determine an orientation of a face of a user relative to the system based on the sensor data. IN addition, the processor may control rotation of the animated face on the display based on the orientation of the system and the orientation of the face of the user to maintain a consistent orientation of the animated face on the display relative to the face of the user.

Inventors: FODOR, JR.; Gregory (Campbell, CA), KALIN; Robert (Catskill, NY), CHATTERTON; Geoffrey W. (Southport, ME)

Applicant: DOPPLE WORKS, INC. (Catskill, NY)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This patent application claims the benefit of and priority to U.S. Provisional Application No. 63/559,099 filed Feb. 28, 2024, titled “INTERACTIVE ARTIFICIAL INTELLIGENCE SYSTEM,” which is incorporated herein by specific reference in its entirety.

FIELD

[0002] The embodiments discussed in the present disclosure are related to adaptive orientation control for interactive artificial intelligence systems.

BACKGROUND

[0003] Unless otherwise indicated in the present disclosure, the materials described in the present disclosure are not prior art to the claims in the present application and are not admitted to be prior art by inclusion in this section.

[0004] An artificial intelligence (AI) system may be utilized in various applications to engage a user via interactions with an AI agent of the AI system. Examples of the AI agent include, but are not limited to, a virtual assistant, a chatbot, or other types of AI agents. The user may interact with the AI agent using different formats such as a text-based format, an audio-based format, an image-based format, or any other appropriate format or combination of formats. For example, the AI system may include natural language processing or speech recognition to process, understand, and respond to voice commands (e.g., audio-based format) or written commands (e.g., text-based format). Such processes allow the AI system to interact with the user via the AI agent more naturally.

[0005] The AI agent may be displayed as a visual representation on a display to create a more engaging and natural interaction experience for the user. For example, the AI agent may be displayed as an animated entity or part of an animated entity. However, some AI systems may display the AI agent in ways that hinder a natural feeling interaction. For example, these AI systems may constrain how the AI agent is visually represented or positioned such that the user may be unable to maintain consistent visual engagement with the AI agent (e.g., the visual representation of the AI agent).

[0006] Some AI systems may be hard for certain users to use. For example, some AI systems may be difficult for children to use because children may not be able to provide input as text and/or children may not be able to naturally engage with these AI systems using voice commands.

[0007] The subject matter claimed in the present disclosure is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some embodiments described in the present disclosure may be practiced.

SUMMARY

[0008] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential characteristics of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0009] A system may include a sensor, a display, and a processor. The sensor may generate sensor data based on a factor of the system. The display may display an animated face of an artificial intelligence (AI) agent. The processor may be coupled to the sensor and the display. The processor may determine an orientation of the system based on the sensor data. The processor may also

determine an orientation of a face of a user relative to the system based on the sensor data. In addition, the processor may control rotation of the animated face on the display based on the orientation of the system and the orientation of the face of the user to maintain a consistent orientation of the animated face on the display relative to the face of the user.

[0010] A system may include a sensor, a display, a housing, and a processor. The sensor may generate sensor data based on a factor of the system. The display may display an animated face of an AI agent such that the animated face occupies an entirety of a viewable area of the display. The animated face may include a facial feature. The housing may house the display. The housing may define the viewable area of the display. The processor may be coupled to the sensor and the display. The processor may determine an orientation of the system based on the sensor data. The processor may also determine an orientation of a face of a user relative to the system based on the sensor data. In addition, the processor may position the animated face on the display based on the orientation of the system and the orientation of the face of the user to maintain a consistent orientation of the animated face on the display relative to the face of the user. Further, the processor may identify an output of the system. The processor may animate the facial feature of the animated face based on the identified output of the system such that the facial feature is synchronized with the output of the system.

[0011] The object and advantages of the embodiments will be realized and achieved at least by the elements, features, and combinations particularly pointed out in the claims. Both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Example embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0013] FIG. 1 illustrates an example environment in which an interactive AI system may be implemented;

[0014] FIGS. 2A and 2B illustrate a front perspective view and a rear perspective view of an example of the interactive AI system of FIG. 1;

[0015] FIGS. 3A and 3B illustrate an example user interacting with the example interactive AI system of FIGS. 2A and 2B in different orientations;

[0016] FIG. 4 illustrates a front perspective view of another example of the interactive AI system of FIG. 1;

[0017] FIGS. 5A and 5B illustrate an example user interacting with the example interactive AI system of FIG. 4 in different orientations;

[0018] FIG. 6 illustrates a block diagram of an example computing device that may be implemented in the environment of FIG. 1,

[0019] all according to at least one embodiment described in the present disclosure.

DETAILED DESCRIPTION

[0020] An AI system may generate output based on input received from the user (referred to generally herein as “user input”), local data (e.g., data stored on the AI system), remote data (e.g., data stored remote from the AI system), or some combination thereof. For example, the user input may include a question, and the AI system may process the user input based on the local data or the remote data to determine and provide, via the AI agent, answers to the question.

[0021] In some embodiments, the user input may include additional data (e.g., additional context) and/or specific settings for the AI system. For example, the user input may indicate that a question involves a specific subject and that the AI system is to limit the answer to that specific subject. The

AI system may process the additional data or the specific settings along with the local data and/or the remote data to refine the output. For example, the AI system may process the additional data, the specific settings, the local data, or the remote data to limit the answer to the question to the specific subject. Accordingly, the AI system may provide a user with means to interact with data and/or the AI agent of the AI system.

[0022] However, some AI systems may lack social presence (e.g., the AI agent is not able to maintain engagement with the user) such that the user does not interact naturally with the AI agent. For example, some AI systems may display the AI agent as a simple icon or as a disembodied being (e.g., a being that is embedded within a digital border), which does not create an intuitive and natural interface. As another example, some AI systems may display the AI agent using a display that breaks an illusion of face-to-face communication due to a shape of the display or an orientation of the AI agent on the display relative to the system. In other words, some AI systems may not create the feeling that the user is speaking directly with the AI agent face-to-face in unmediated communication.

[0023] There are opportunities to improve how the AI agent is displayed to create an intuitive and natural interface for the user, which may increase user engagement and overall utility of the AI system.

[0024] The present disclosure provides an AI system that creates an intuitive and natural interface between the user and the AI system. The AI system of the present disclosure enables natural interactions between the AI system and the user. Additionally, the AI system displays the AI agent in a manner that causes the AI agent to form part of a computing device that is implementing the AI system and not merely as a disembodied being.

[0025] The AI system may display the AI agent as an animated face on a surface of the device via a display. In some embodiments, the display may include a shape that is the same or similar to a shape of the face of the AI agent to make the face of the AI agent appear as if to form part of the computing device. Additionally, as the AI system (e.g., the computing device) is rotated, the AI system may rotate the AI agent on the display to maintain orientation of the animated face of the AI agent relative to eyes of the user rather than an orientation of the AI system.

[0026] The AI system may include a sensor, a display, or a processor. The sensor may generate sensor data based on a factor of the system (e.g., based on movement or a location of the AI system). The display may display an animated face of the AI agent such that the AI agent forms part of the computing device implementing the AI system. The processor may determine an orientation of the AI system (e.g., the computing device) based on the sensor data. In addition, the processor may determine an orientation of a face of the user relative to the AI system (e.g., relative to the display) based on the sensor data. Further, the processor may control rotation of the animated face on the display based on the orientation of the AI system, the orientation of the face of the user, or both. Additionally, the processor may control the rotation of the animated face to maintain a consistent orientation of the animated face of the AI agent relative to the face of the user.

[0027] The AI system may include a housing that permits the user to hold the AI system at a variety of angles or orientations. In addition, the processor may control rotation of the animated face to maintain a consistent orientation relative to the user to permit the user to hold (e.g., position) the housing and/or any part of the AI system based on a preference of the user rather than constraints of the AI system. For example, the user may hold a handle and/or an upper body of the AI system in their hands in a preferred (e.g., comfortable or convenient) position rather than keep the AI system in a particular orientation.

[0028] Additionally or alternatively, the AI system may include an input device (e.g., a camera) configured to capture input data. For example, the camera may capture image data representative of a scene. The processor may control orientation of the input data based on the orientation of the user, the direction of gravity, or both. In particular, the processor may permit the orientation of the input data to be based on the orientation of the user, the direction of gravity, or both rather than the

orientation of the input data being fixed relative to the input device or the AI system. For example, the processor may cause an orientation of a view of the camera to be based on the orientation of the user, the direction of gravity, or both rather than the orientation of the view being fixed relative to the camera or the AI system. As another example, the processor may process the image data to reorient the image data relative to the orientation of the user, the direction of gravity, or both.

[0029] The AI system described in the present disclosure may permit the user to naturally interact with the AI agent while the user holds the computing device implementing the AI system at a variety of angles or orientations compared to an AI system that displays the AI agent in a fixed orientation. Further, the AI system described in the present disclosure displays the AI agent in a manner that fuses or embeds the animated face of the AI agent in the computing device implementing the AI system such that the AI agent appears as a face of the computing device. Therefore, the AI agent includes a social presence that creates a feeling of face-to-face communication between the user and the AI system (e.g., the computing device) rather than causing the user to feel like they are interacting with a disembodied being.

[0030] The AI system described in the present disclosure permits the user to hold the AI system based on the preference of the user and to capture input data based on the orientation of the user, the direction of gravity, or both. For example, the AI system may permit the user to capture a selfie using the camera and the selfie can be oriented based on the orientation of the user, the direction of gravity, or both rather than the orientation of the AI system.

[0031] These and other embodiments of the present disclosure will be explained with reference to the accompanying figures. It is to be understood that the figures are diagrammatic and schematic representations of such example embodiments, and are not limiting, nor are they necessarily drawn to scale. In the figures, features with like numbers indicate like structure and function unless described otherwise.

[0032] FIG. 1 illustrates an example environment **100** in which an interactive AI system **102** may be implemented, in accordance with one or more embodiments of the present disclosure. The environment **100** may include the interactive AI system **102**, a computing device **104**, a network **106**, and/or a cloud computing device **108**. The interactive AI system **102** may include or be implemented on a computing device, an example of which is shown and described below in relation to FIG. 6.

[0033] The interactive AI system **102** (referred to herein as the AI system **102**) may include a display **110** configured to display an AI agent **124**. The AI system **102** may also include a speaker **112**, a processor **114**, a memory **116**, an input device **118**, or one or more sensors **120**. The processor **114** may be coupled to the display **110**, the speaker **112**, the memory **116**, the input device **118**, or the sensors **120**.

[0034] The AI system **102** may permit a user **136** to interact with AI or the AI agent **124** in a manner that creates the illusion of face-to-face communication between the user **136** and the AI system **102**. For example, the AI system **102** may include a hand-held device that permits the user **136** to provide user input to the AI system **102** in a user-friendly way. Examples of hand-held versions of the AI system **102** are shown and described in more detail below in relation to FIGS. 2A, 2B, and 4. The AI agent **124** may be displayed so as to create an illusion that the user **136** is interacting with the AI system **102** directly as a specific entity rather than merely a machine or a disembodied being. Further, the AI system **102** may generate output based on the user input and animate the AI agent **124** such that the AI agent **124** is synchronized and appears to be speaking the output.

[0035] The user **136** may interact with the AI agent **124** or the AI system **102** via the input device **118**. Additionally or alternatively, the user **136** may interact with the AI agent **124** via the display **110**. Interacting with the AI agent **124** of the present disclosure may include interacting with the visual representation of the AI agent **124** (e.g., the animated face), the input device **118**, or any other appropriate aspect of the AI system **102**.

[0036] The user **136** may provide the user input to control the AI system **102** or interact with the AI agent **124**. For example, the user **136** may press and hold a button **144** of the input device **118** while speaking to provide the user input as a voice input. As another example, the user **136** may release the button **144** to signal the end of the user input. As another example, the button **144** may mute/unmute the AI system **102**, adjust a volume of the AI system **102**, or activate different modes of operation of the AI system **102**.

[0037] In some embodiments, the input device **118** may capture the user input and generate input data **128** representative of the user input. For example, the input device **118** may receive the user input as voice commands via a microphone **146** and the microphone **146** may generate the input data **128** as a waveform representative of the voice commands. In these and other embodiments, the input device **118** may capture the user input and the processor **114** may generate the input data **128** representative of the user input. For example, the input device **118** may receive the user input as touch commands via the display **110** and/or the button **144** and the processor **114** may generate the input data **128** accordingly. In some embodiments, the button **144** may include a capacitance device configured to detect a touch of the user **136**.

[0038] The input device **118** may include a camera **122** that captures image data representative of the user **136** or of the environment **100** (e.g., a scene) within a view of the camera **122**. In addition, the input data **128** may include the image data. In some embodiments, the user **136** may direct the camera **122** at a document, a picture, or any other appropriate object to permit the camera **122** to capture the image data representative of the object. For example, the user **136** may direct the camera **122** at a page of a book and the input data **128** may be representative of words on the page. As another example, the user **136** may direct the camera **122** at a picture within the environment **100** and the input data **128** may be representative of the picture.

[0039] In some embodiments, the camera **122** may capture the image data with an orientation that is based on the orientation of the user **136**, a direction of gravity, or both. For example, the camera **122** may receive instructions from the processor **114** indicating an orientation at which the camera **122** is to capture the image data (e.g., the instructions may indicate the orientation of the user **136**, the direction of gravity, or both). As another example, the camera **122** may include a sensor (not shown) configured to detect the direction of gravity.

[0040] In some embodiments, the processor **114** may control an orientation of a view of the camera **122** based on the orientation of the user **136**, the direction of gravity, or both. In other embodiments, the camera **122** may capture the image data based on a fixed orientation of the view of the camera **122** and the processor **114** may process the image data to reorient it based on the orientation of the user **136**, the direction of gravity, or both.

[0041] The microphone **146** may capture the voice commands spoken by the user **136** or sounds in the environment **100**. Additionally, the input data **128** may include audio data representative of the voice commands or the sounds in the environment **100**. In some embodiments, the user **136** may speak one or more phrases which are captured by the microphone **146** as the audio data. In some embodiments, the microphone **146** or the processor **114** may generate the input data **128** based on the audio data. For example, the user **136** may speak a voice command to search through a particular database, perform a calculation, set a reminder, or any other appropriate command and the processor **114** may generate the input data **128** representative of the command.

[0042] In some embodiments, the user input received via different devices may be combined in the input data **128**. For example, the user **136** may direct the camera **122** at an object and speak about what the camera **122** is directed at and the processor **114** may associate the image data and the audio data in the input data **128**.

[0043] The display **110** may include a touch sensitive surface configured to capture touch input from the user **136** and the input device **118** may generate the input data **128** based on the touch input. For example, the user **136** may touch a portion of the display **110** to move the AI agent **124**.

[0044] The processor **114** may cause the display **110** to display the AI agent **124** as an animated

face. In other words, the display **110** may display the animated face of the AI agent **124**. The AI agent **124** (e.g., the animated face) may include facial features. For example, the AI agent **124** may include a nose, a skin tone, hair, facial hair, a mouth, eyes, eyebrows, a forehead, a chin, cheeks, or any other appropriate facial feature. In some embodiments, the AI agent **124** may include different accessories such as earrings, makeup, or other appropriate accessories.

[0045] In some embodiments, the processor **114** may cause the AI agent **124** to be displayed to represent a human. In other embodiments, the processor **114** may cause the AI agent **124** to be displayed to represent a non-human such as an animal (e.g., a dog, a cat, or a bear) or a fantasy creature (e.g., a dragon, an elf, or an angel).

[0046] In some embodiments, the AI agent **124** may be displayed such that the animated face occupies an entirety of a viewable area of the display **110**. Accordingly, the AI agent **124** may cause the display **110** to appear as the face of the AI system **102** rather than the AI agent **124** appearing to merely be an avatar within a screen or an icon on a display. The viewable area of the display **110** is discussed in more detail below in relation to FIGS. 2A, 2B, and 4.

[0047] The processor **114** may animate the AI agent **124** to indicate information to the user **136**. For example, the processor **114** may animate the AI agent **124** such that eyes of the AI agent **124** blink or look around the environment **100** to indicate that the AI agent **124** is active and ready to interact with the user **136**. As another example, the processor **114** may cause the eyes of the AI agent **124** to glance in directions to indicate to the user **136** that the AI system **102** detected something of interest in those directions. As another example, the processor **114** may animate the AI agent **124** such that the eyes of the AI agent **124** follow eyes of the user **136** to maintain eye contact as described in more detail below.

[0048] In some embodiments, the AI system **102** may receive viseme data **126**, operation data **142**, or any other appropriate data from the computing device **104** or the cloud computing device **108**. For example, the computing device **104** may be located proximate to the AI system **102** and the AI system **102** may receive the viseme data **126** or the operation data **142** via a wired connection. As another example, the AI system **102** may receive the viseme data **126** or the operation data **142** from the cloud computing device **108** either directly via the network **106** or indirectly via the network **106** and a wired connection to the computing device **104**.

[0049] The operation data **142** may include any appropriate information to allow the AI system **102** to perform operations described in the present disclosure. For example, the operation data **142** may include data configured to permit the AI system **102** to perform object detection, process information, identify content of user input, generate output, or any other appropriate function. As another example, the operation data **142** may permit the AI system **102** to operate as if including an AI model or other processing model.

[0050] The memory **116** may store the viseme data **126**, the input data **128**, output data **130**, sensor data **132**, or the operation data **142** for use by the AI system **102** (e.g., the processor **114**). The viseme data **126** may represent visual positions and movements of facial features corresponding to different speech sounds or various words. The output data **130** may represent the output that is generated by the AI system **102**. In some embodiments, the output data **130** may represent audio samples, phonemes, words, phrases, and other speech-related information used for speech synthesis. In these and other embodiments, the output data **130** may permit the AI system **102** to provide natural-sounding speech output for the AI agent **124** via the speaker **112** or any other appropriate type of output.

[0051] The processor **114** may generate the output data **130** based on the input data **128**, the sensor data **132**, the viseme data **126**, the operation data **142**, or any other appropriate data stored in the memory **116**. Alternatively, the processor **114** may generate the output data **130** based on the viseme data **126**, the operation data **142**, input data **128**, the sensor data **132**, the viseme data **126**, the operation data **142**, or any other appropriate data stored in a memory **139** of the computing device **104** or a memory **138** of the cloud computing device **108**.

[0052] The processor **114** may cause the output to be provided to the user **136** as audio output via the speaker **112**, as haptic output via a housing, as visual output via the display **110**, or via any other appropriate method. Additionally, the processor **114** may animate the AI agent **124** to be synchronized with the output. For example, the processor **114** may animate the facial features of the AI agent **124** to correspond to the audio output of the AI system **102** to make the AI agent **124** appear as if it is speaking the audio output.

[0053] In some embodiments, the processor **114** may identify words that form at least part of the output and use the viseme data **126** to animate the AI agent **124** in synchronization with the words generated using the speaker **112**. For example, the output data **130** may include an audio waveform and the processor **114** may estimate poses for the facial features of the AI agent **124** using the viseme data **126** to correspond to the audio waveform when played on the speaker **112**. In some embodiments, the processor **114** may implement a neural network to estimate the poses of the facial features of the AI agent **124**. In these and other embodiments, the processor **114** may estimate the poses as control point settings, vertex morphs, or any other appropriate method to describe poses when animating a digital character.

[0054] The sensors **120** (generally referred to in the present disclosure as the sensor **120**) may generate the sensor data **132** based on factors of the AI system **102**. The sensor data **132** may be representative of the factors of the AI system **102**. For example, the sensor data **132** may be representative of the orientation, a location, or movement of the AI system **102**. The sensor **120** may include an accelerometer, an inertial measurement unit (IMU), a magnetometer, a gyroscope, a magnetic compass, or a light detection and ranging sensor. The sensor **120** may measure and/or report acceleration, orientation, angular rates, and other gravitational forces of the AI system **102**. In some embodiments, the sensor **120** may include a camera (e.g., the camera **122**) configured to capture image data representative of a location or orientation of a face, facial features, or other parts of the face (e.g., eyes) of the user **136**.

[0055] The processor **114** may determine an orientation of the AI system **102** or the display **110** based on the sensor data **132**. For example, the processor **114** may process the sensor data **132** to determine the orientation of the AI system **102**. Additionally or alternatively, the processor **114** may determine an orientation of the face of the user **136** (e.g., eyes of the user **136**) relative to the AI system **102** (e.g., the display **110**) based on the sensor data **132**. For example, the processor **114** may perform object detection in images captured by the camera of the sensor **120** to locate the eyes of the user **136**. As another example, the sensor data **132** may include LiDAR data and the processor **114** may process the LiDAR data to determine a location of facial features including the eyes of the user **136**.

[0056] The processor **114** may control rotation of the AI agent **124** within the display **110** based on the orientation of the system, the orientation of the face of the user **136**, or both. In particular, the processor **114** may orient the animated face of the AI agent **124** on the display **110** to maintain a consistent orientation of the animated face of the AI agent **124** relative to the user **136**. For instance, the processor **114** may orient the animated face of the AI agent **124** on the display **110** to maintain eye contact between the AI agent **124** and the user **136** regardless of the orientation of the AI system **102**.

[0057] In some embodiments, the processor **114** may control the rotation of the AI agent **124** on the display **110** to maintain a consistent orientation of the animated face (e.g., the facial features) on the display **110** with respect to the face of the user **136**. In other words, the processor **114** may control the rotation of the AI agent **124** on the display **110** such that the orientation of the AI agent **124** is aligned with the orientation of the face of the user **136** regardless of the orientation, movement, or both of the AI system **102**. Examples of the orientation of the AI agent being aligned with the orientation of the face of the user **136** are shown and described in more detail below in relation to FIGS. 3A, 3B, 5A, and 5B.

[0058] The processor **114** may control the rotation of the animated face of the AI agent **124** on the

display **110** to permit the user **136** to position the AI system **102** in different positions (e.g., different desired positions) or orientations based on a preference of the user **136** rather than factors of the display **110** or the AI system **102** (e.g., based on the desired positions rather than a constraint of the AI system **102**).

[0059] The processor **114** may control placement or angles of the facial features of the AI agent **124** to maintain the eye contact between the user **136** and the AI agent **124**. The processor **114** may track movement of the eyes of the user **136** based on the image data captured by the camera of the sensor **120** (e.g., perform face tracking using image data captured by a front facing camera (not shown)). Additionally, the processor **114** may control a gaze of the eyes of the AI agent **124** based on the movement of the eyes of the user **136**. For example, if the user **136** looks to their left, the processor **114** may animate the eyes of the AI agent **124** to also look to the left of the user **136**.

[0060] Alternatively, the processor **114** may control the placement or angles of the eyes of the AI agent **124** to cause the AI agent **124** to appear to be looking at an object the user **136** is looking at. For example, the eyes of the user **136** may move up to look at a ceiling and the processor **114** may cause the eyes of the AI agent **124** to also look up at the ceiling.

[0061] The processor **114** may animate the AI agent **124** based on operations performed by the AI system **102**. For example, the processor **114** may cause a mouth of the AI agent **124** to disappear to indicate that the AI system **102** has been muted. As another example, the processor **114** may cause the eyes of the AI agent **124** to close to indicate that the AI system **102** is in a sleep mode. As yet another example, the processor **114** may cause the eyes of the AI agent **124** to wink to indicate that the AI agent **124** has a secret to communicate to the user **136**.

[0062] The processor **114**, when the user **136** is providing the user input, may animate the AI agent **124** to show that the user **136** has the attention of the AI agent **124**. Additionally or alternatively, the processor **114** may cause the AI agent **124** to move to create the illusion that the AI agent **124** is thinking while performing calculations or searching databases.

[0063] The processor **114** may cause the AI system **102** to transition between modes based on the user input. In some embodiments, the AI system **102** may operate in a sleep mode or an active mode. In the sleep mode, the display **110** may be turned off or in a standby mode. In addition, in the sleep mode, the microphone **146** may capture audio input and the processor **114** may determine if the audio input includes a wake command or a wake word. For example, the processor **114** may determine if the audio input includes a name of the AI system **102** or a greeting for the AI system **102**.

[0064] If the audio input includes a wake command or a wake word, the processor **114** may cause the display **110** to transition from the off or standby mode to the active mode. In addition, the processor **114** may cause the animated face of the AI agent **124** to be displayed on the display **110**. In some embodiments, the wake command may be predetermined. In other embodiments, the wake command may be determined and/or set by the user **136**.

[0065] The processor **114** may cause the AI system **102** to transition from the sleep mode to the active mode based on the user input received via the button **144** or the display **110**. For example, the user **136** may press the button **144** or touch the display **110** to cause the AI agent **124** to wake up (e.g., cause the AI system **102** to transition to the active mode) and begin a conversation with the user **136**.

[0066] In some embodiments, the processor **114** may animate the facial features of the AI agent **124** such that the AI agent **124** appears to show or create an illusion of having different emotions. For example, the one or more facial features of the AI agent **124** may be animated to show happiness (e.g., a happy face), sadness (e.g., a sad face), excitement (e.g., an excited face), or any other appropriate emotion or face.

[0067] In some embodiments, the processor **114** may determine an emotion corresponding to the output of the AI system **102**. In these and other embodiments, the processor **114** may animate the AI agent **124** to show the emotion corresponding to the output of the AI system **102**. For example,

the processor **114** may determine the user **136** is happy by processing image data representative of the user **136** and the processor **114** may animate the AI agent **124** to reflect happiness.

[0068] In some embodiments, the processor **114** may animate the AI agent **124** to appear to react to the user **136** touching the display **110**. For example, when the user **136** touches a certain part of the display **110**, the processor **114** may animate the AI agent **124** as if reacting to the touch.

[0069] In some embodiments, the processor **114** may be configured to cause the display **110** to display the AI agent **124** with different faces or facial features. In these and other embodiments, the different faces or facial features may be displayed based on different functions and/or operations being performed by the AI system **102**. For example, the processor **114** may cause different faces to be displayed for different applications being run by the AI system **102**. Additionally or alternatively, the processor **114** may cause different faces or facial features to be displayed based on an identity of the user **136**. In other words, the processor **114** may cause different faces or facial features to be displayed for different users. In some embodiments, the user **136** may select the face or facial features of the AI agent **124** that are displayed.

[0070] In some embodiments, the AI system **102** may be associated with a dock (not shown). In some embodiments, the dock may receive the AI system **102** to stabilize, charge, or secure the AI system **102**. For example, the AI system **102** may include a power source (not shown) such as a rechargeable battery and the dock may charge the power source.

[0071] The network **106** may include any communication network configured for communication of signals between any of the components (e.g., **102**, **104**, or **108**) of the environment **100**. The network **106** may be wired or wireless. The network **106** may have numerous configurations including a star configuration, a token ring configuration, or another suitable configuration. Furthermore, the network **106** may include a local area network (LAN), a wide area network (WAN) (e.g., the Internet), and/or other interconnected data paths across which multiple devices may communicate. In some embodiments, the network **106** may include a peer-to-peer network. The network **106** may also be coupled to or include portions of a telecommunications network that may enable communication of data in a variety of different communication protocols.

[0072] In some embodiments, the network **106** includes or is configured to include a BLUETOOTH® communication network, a Z-Wave® communication network, an Insteon® communication network, an EnOcean® communication network, a wireless fidelity (Wi-Fi) communication network, a ZigBee communication network, a HomePlug communication network, a Power-line Communication (PLC) communication network, a message queue telemetry transport (MQTT) communication network, a MQTT-sensor (MQTT-S) communication network, a constrained application protocol (CoAP) communication network, a representative state transfer application protocol interface (REST API) communication network, an extensible messaging and presence protocol (XMPP) communication network, a cellular communications network, any similar communication networks, or any combination thereof for sending and receiving data. The data communicated in the network **106** may include data communicated via short messaging service (SMS), multimedia messaging service (MMS), hypertext transfer protocol (HTTP), direct data connection, wireless application protocol (WAP), e-mail, smart energy profile (SEP), ECHONET Lite, OpenADR, or any other protocol that may be implemented with the AI system **102**, the computing device **104**, or the cloud computing device **108**.

[0073] The processor **114** may include a central processing unit (CPU), a microprocessor (μ P), a microcontroller (μ C), a digital signal processor (DSP), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or any combination thereof. The processor **114** may be configured to execute computer instructions that, when executed, cause the processor **114** or the AI system **102**, to perform or control performance of one or more of the operations described herein with respect to operation of the AI system **102**. The processor **114** may be implemented using a combination of hardware and software. In the present disclosure, operations described as being performed by the processor **114** or the AI system **102** may include operations that the

processor **114** or the AI system **102** directs a corresponding system to perform.

[0074] The memories **116**, **138**, **139** may include storage mediums for storing data and instructions. Examples of memory devices that may be used include, but are not limited to random access memory (RAM), persistent or non-volatile storage such as read-only memory (ROM), NAND flash memory, solid state drives (SSDs), hard disk drives (HDDs), optical discs (e.g. CD-ROM, DVD-ROM), or magnetic tapes. The memories **116**, **138**, **139** may include volatile memory (e.g. RAM) for temporary storage and non-volatile memory (e.g. ROM, flash memory) for persistent storage. The memories **116**, **138**, **139** may store computer instructions that may be executed by the processor **114**, the computing device **104**, or the cloud computing device **108** to perform or control performance of one or more of the operations described herein with respect to operation of the AI system **102**. In addition, the memories **116**, **138**, **139** may store the viseme data **126**, the input data **128**, the output data **130**, the sensor data **132**, the operation data **142** persistently and/or at least temporarily.

[0075] FIGS. 2A and 2B illustrate a front perspective view and a rear perspective view of an example of the AI system **102** of FIG. 1, in accordance with one or more embodiments of the present disclosure.

[0076] The AI system **102** may include a housing **248** that houses or contains components of the AI system **102**. The housing **248** may at least partially house the processor **114** (not shown in FIGS. 2A and 2B), the memory **116** (not shown in FIGS. 2A and 2B), the sensor **120** (not shown in FIGS. 2A and 2B), the input device **118**, or the display **110**. In some embodiments, the housing **248** may include an upper body **251** or a handle **250**. The upper body **251** may house the display **110**.

[0077] As shown in FIGS. 2A and 2B, at least a portion of the upper body **251** includes a substantially circular shape. In some embodiments, the upper body **251** is shaped substantially circular, such that the upper body **251** looks substantially the same in various orientations of the AI system **102**. Although illustrated as including a circular shape in FIGS. 2A and 2B, the upper body **251** may include any suitable shape. For example, the upper body **251** may include a triangular shape, a diamond shape, a rectangular shape, a square shape, or any other appropriate shape.

[0078] In some embodiments, the upper body **251** may define a viewable area **252** of the display **110**. Although illustrated as including a circular shape in FIGS. 2A and 2B, the viewable area **252** may include any suitable shape such as an oval. Alternatively, the viewable area **252** may include a non-circular shape. The viewable area **252** may include a portion of the display **110** that is exposed or visible to the user **136**. The AI agent **124** may be displayed within the viewable area **252** such that the animated face **254** of the AI agent **124** occupies an entirety of the viewable area **252**. In addition, the AI agent **124** may be displayed within the viewable area **252** such that the upper body **251** forms a portion or a boundary of a head of the AI agent **124**. As shown in FIG. 2A, the AI agent **124** includes eyes, a mouth, and a face.

[0079] The display **110** may display the animated face **254** such that the animated face **254** is substantially centered around a point of the display **110**. In some embodiments, the point may include a center axis of the display **110**, a center axis of the viewable area **252**, or both. In other embodiments, the point may include any point of the display **110**. For example, the housing **248** may define the viewable area **252** off center of the display **110** and the point may include the center axis of the viewable area **252**.

[0080] The processor **114** may control the rotation of the animated face **254** such that the animated face **254** rotates around the point of the display **110**. In some embodiments, the processor **114** may cause the animated face **254** to rotate around the point of the display **110** to maintain substantially consistent distances between individual facial features of the animated face **254** and a boundary of the viewable area **252**. For example, the processor **114** may cause the animated face **254** to rotate around the point of the display **110** to maintain a substantially consistent distance between the eyes, the mouth, or any other appropriate facial feature of the animated face **254** and the boundary of the viewable area **252** when the AI system **102** is at different orientations. The boundary of the

viewable area **252** may be defined by edges of the display **110**, the housing **248**, or both. In these and other embodiments, the processor **114** may cause the animated face **254** to rotate around the point of the display **110** to maintain a substantially consistent distance between the eyes, the mouth, or any other appropriate facial feature of the animated face **254** and edges of the housing **248** when the AI system **102** is at different orientations. For example, the processor **114** may cause the animated face **254** to rotate around the point of the display **110** to maintain a substantially consistent distance between the eyes, the mouth, or any other appropriate facial feature of the animated face **254** and the edges of the housing **248** when the AI system **102** is at different orientations

[0081] In some embodiments, the display **110** may substantially form a front portion or part of a surface of the AI system **102** such that the animated face **254** appears to be embodied by the AI system **102**. The display **110** shown in FIG. 2A includes a circular shape. However, the display **110** may include a non-circular shape but the upper body **251** may define the viewable area **252** as including the substantially circular shape. In some embodiments, the animated face **254** of the AI agent **124** may be displayed (e.g., depicted) on the display **110** using a color scheme for the skin tone of the AI agent **124**. For example, the animated face **254** of the AI agent **124** may be represented using a single color (e.g., blue).

[0082] The input device **118** is shown in FIGS. 2A and 2B as including multiple buttons **144a-c**. Each of the buttons **144a-c** may correspond to different functions (e.g., push to talk, push to wake, or push to capture image data). Alternatively, two or more of the buttons **144a-c** may correspond to the same function. For example, the button **144a-b** may both correspond to push to talk. In some embodiments, a back surface **253** of the AI system **102** may include a capacitive touch surface to act as a button or other device to receive the user input.

[0083] The button **144c** is shown in FIGS. 2A and 2B as being positioned on a rim or an edge of the upper body **251** for easy access. In embodiments with the handle **250**, the buttons **144a-b** may be located on a front or a back of the handle **250**. The exact placement and number of buttons may vary based on the specific form factor and ergonomics of the AI system **102**. For example, the handle **250** may be omitted and the input device **118** may only include the button **144c**. The buttons **144a-c** may include a trigger, an analog stick, a touchpad, a button, a squeeze grip, a D-pad, or some combination thereof.

[0084] The upper body **251** may protect the display **110** from being damaged during transport or use of the AI system **102**. For example, sides of the display **110** may be covered by the upper body **251** to prevent the sides of the display **110** or other portions of the display **110** from being damaged. Additionally or alternatively, the upper body **251** may be thicker than the display **110** to cause the upper body **251** to contact a ground or other surface more than the display **110** if the AI system **102** is dropped, for example.

[0085] In some embodiments, the handle **250** (e.g., lower body) may be used by the user **136** to hold the AI system **102**. For example, the user **136** may hold the handle **250** in their hand. In some embodiments, the handle **250** may be shaped in any suitable shape for the user **136** to hold. For example, the handle **250** may include a cylindrical shape. Additionally or alternatively, the upper body **251** may be used by the user **136** to hold the AI system **102**.

[0086] In some embodiments, the upper body **251** may be connected to the handle **250**. In some embodiments, the handle **250** may stem from a side of the upper body **251**. The handle **250** may include any suitable shape to act as a handle. For example, the handle **250** may have a cylindrical, cubic, cuboid, and/or any other suitable shape.

[0087] In embodiments that include the handle **250**, the upper body **251** may include a ring that sits atop the handle **250** to cause the AI system **102** to appear like a magnifying glass. In other embodiments, the handle **250** may be omitted and the AI system **102** may be shaped like a disc. In some embodiments, the AI system **102** may be large enough to be held with two hands, and in other embodiments the AI system **102** may be small enough to be held with one hand.

[0088] As shown in FIG. 2B, the camera **122** is positioned on an opposite side as the display **110**. Additionally or alternatively, the AI system **102** may include a camera positioned on the same side as the display **110**. In some embodiments, the AI system **102** may include multiple cameras positioned on various portions of the AI system **102**—e.g., a first camera positioned on the same side as the display **110** and a second camera positioned on a side opposite to the display **110**. The AI system **102** may include a power source (not shown). In some embodiments, the power source may be housed in the handle **250**.

[0089] In some embodiments, the AI system **102** may include a light source **261** configured to assist the camera **122** capture image data. For example, the light source **261** may operate in response to the camera **122** not detecting enough light to properly capture the image data.

[0090] As shown in FIG. 2B, the AI system **102** includes two microphones **146a-b** to capture audio data. One or more of the microphones **146a-b** may be omitted. Additionally or alternatively, more than two microphones **146a-b** may be used to capture audio data.

[0091] FIGS. 3A and 3B illustrate an example user **136** interacting with the AI system **102** of FIGS. 2A and 2B in different orientations, in accordance with one or more embodiments of the present disclosure. As shown in FIG. 3A, the user **136** is holding the AI system **102** such that the handle **250** extends generally up in the drawing and the handle **250** is in the left hand of the user **136**. As shown in FIG. 3B, the user **136** is now holding the AI system **102** such that the handle **250** extends generally to the right and down in the drawing and the handle **250** is in the right hand of the user **136**. The user **136** may move the AI system **102** or rotate the AI system **102** (e.g., in a clockwise or counterclockwise direction) to change the orientation of the AI system **102**. For example, as shown in FIGS. 3A and 3B, the user **136** switched hands that are holding the handle **250** to rotate and change the orientation of the AI system **102**.

[0092] Despite the orientation of the AI system **102** changing between FIGS. 3A and 3B (as shown by the handle **250** extending in different directions), the orientation of the animated face **254** of the AI agent **124** is maintained relative to the user **136**. For example, the mouth and eyes of the animated face **254** are positioned in both orientations of the AI system **102** so as to have eye contact with the user **136** or to match an orientation of the user **136**. Accordingly, regardless of the orientation of the AI system **102**, the animated face **254** of the AI agent **124** is displayed on the display **110** such that the AI agent **124** appears in a same orientation as the user **136**.

[0093] FIG. 4 illustrates a front perspective view of another example of the AI system **102** of FIG. 1, in accordance with one or more embodiments of the present disclosure. The example AI system **102** shown in FIG. 4 includes a smaller form factor than the example of the AI system **102** shown in FIGS. 2A and 2B. For example, the example AI system **102** shown in FIG. 4 includes a smartwatch that can be worn on a wrist of the user **136** or placed in a pocket of the user. Form factor refers to a size, a shape, or a physical design of the AI system **102** and how the AI system **102** fits, interacts, or interfaces with other devices, such as hands of the user **136**.

[0094] The example AI system **102**, as shown in FIG. 4, includes a housing **448**. The housing **448** may operate the same as or similar to the upper body **251** discussed above in relation to FIGS. 2A and 2B. For example, the housing **448** may define the viewable area **252** of the display **110**. As another example, the housing **448** may house various components of the example AI system **102**.

[0095] The animated face **254** may be displayed on the display **110** such that the orientation of the animated face **254** is maintained relative to the user as described in more detail below in relation to FIGS. 5A and 5B.

[0096] The AI system **102** may be sized to permit smaller hands (e.g., the hands of a child) to interact with the AI system **102** or to permit the AI system **102** to fit on a wrist or in a pocket of the user **136**. For example, the AI system **102** may be sized similar to a watch face, a smartphone, a tablet, a laptop, among others. A watch strap (not shown) may be connected to the AI system **102**. In some embodiments, the AI system **102** may be placed or connected to a dock **434** to stabilize, charge, or perform any other appropriate operation.

[0097] FIGS. 5A and 5B illustrate an example user interacting with the example AI system **102** of FIG. **4**, in accordance with one or more embodiments of the present disclosure. FIGS. 5A and 5B illustrate the example AI system **102** in the hands of the user **136** from a perspective of the user **136** (e.g., as if viewed by the user **136**). As shown in FIG. 5A, the user **136** is holding the AI system **102** such that band connecting portions **551** extend generally up and to the right in the drawing and the AI system **102** is in the left hand of the user **136**. As shown in FIG. 5B, the user **136** is now holding the AI system **102** such that the band connecting portions **551** extend generally up and to the left in the drawing and the AI system **102** is in the right hand of the user **136**. The user **136** may move the AI system **102** or rotate the AI system **102** (e.g., in a clockwise or counterclockwise direction) to change the orientation of the AI system **102**. For example, as shown in FIGS. 5A and 5B, the user **136** switched hands that are holding the AI system **102** to rotate and change the orientation of the AI system **102**.

[0098] Despite the orientation of the AI system **102** changing between FIGS. 5A and 5B (as shown by the band connecting portions **551** extending in different directions and the AI system **102** being in different hands of the user **136**), the orientation of the animated face **254** of the AI agent **124** is maintained relative to the perspective of the user **136**. For example, the mouth and eyes of the animated face **254** are positioned in both orientations of the AI system **102** to have eye contact with the user **136** (e.g., appear to look towards or to match an orientation of the user **136**). Accordingly, regardless of the orientation of the AI system **102**, the animated face **254** of the AI agent **124** is displayed on the display **110** such that the AI agent **124** appears in a same orientation as the user **136**.

[0099] FIG. **6** illustrates a block diagram of an example computing system **600** that may be implemented in the environment **100** of FIG. **1**, in accordance with one or more embodiments of the present disclosure. In some embodiments, the computing system **600** may be implemented on the AI system **102**, the computing device **104**, or the cloud computing device **108** of FIG. **1**. The computing system **600** may implement or direct one or more suitable operations described in the present disclosure. For example, the computing system **600** may be configured to control the AI system **102** of FIGS. 1-5B, the computing device **104** of FIG. **1**, or the cloud computing device **108** of FIG. **1**. Additionally, the computing system **600** may obtain and process user input to generate a corresponding output as described in relation to FIG. **1**. The computing system **600** may include a processor **602**, memory **604**, a power source **606**, an input interface **608**, a display **610**, a communication unit **612**, and a sensor **616**. The processor **602**, the memory **604**, the power source **606**, the input interface **608**, the display **610**, the communication unit **612**, or the sensor **616** may be communicatively coupled.

[0100] Generally, the processor **602** may include any suitable special-purpose or general-purpose computer, computing entity, or processing device including various computer hardware or software modules and may be configured to execute instructions stored on any applicable computer-readable storage media. For example, the processor **602** may include a microprocessor, a microcontroller, a parallel processor such as a graphics processing unit (GPU) or tensor processing unit (TPU), a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a Field-Programmable Gate Array (FPGA), or any other digital or analog circuitry configured to interpret and/or to execute program instructions and/or to process data.

[0101] Although illustrated as a single processor in FIG. **6**, it is understood that the processor **602** may include any number of processors distributed across any number of networks or physical locations that are configured to perform individually or collectively any number of operations described herein. In some embodiments, the processor **602** may interpret and/or execute program instructions and/or process data stored in the memory **604**. In some embodiments, the processor **602** may execute the program instructions stored in the memory **604**. For example, in some embodiments, the processor **602** may execute program instructions stored in the memory **604** that are related to task execution such that the computing system **600** may perform or direct the

performance of the operations associated therewith as directed by the instructions.

[0102] The memory **604** may include computer-readable storage media or one or more computer-readable storage mediums for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable storage media may be any available media that may be accessed by a general-purpose or special-purpose computer, such as the processor **602**.

[0103] By way of example, and not limitation, such computer-readable storage media may include non-transitory computer-readable storage media including Random Access Memory (RAM), Read-Only Memory (ROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), Compact Disc Read-Only Memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices, flash memory devices (e.g., solid state memory devices), or any other storage medium which may be used to carry or store particular program code in the form of computer-executable instructions or data structures and which may be accessed by a general-purpose or special-purpose computer. Combinations of the above may also be included within the scope of computer-readable storage media.

[0104] Computer-executable instructions may include, for example, instructions and data configured to cause the processor **602** to perform a certain operation or group of operations as described in this disclosure. In these and other embodiments, the term “non-transitory” as explained in the present disclosure should be construed to exclude only those types of transitory media that were found to fall outside the scope of patentable subject matter in the Federal Circuit decision of *In re Nuijten*, 500 F.3d 1346 (Fed. Cir. 2007). Combinations of the above may also be included within the scope of computer-readable media.

[0105] The power source **606** may include one or more energy storage devices for powering components of the computing system **600**. For example, the power source **606** may include one or more batteries. In some embodiments, the one or more batteries may be removable, replaceable, and/or rechargeable.

[0106] The communication unit **612** may include any component, device, system, or combination thereof that is configured to transmit or receive information over a network. In some embodiments, the communication unit **612** may communicate with other devices at other locations, the same location, or even other components within the same system. For example, the communication unit **612** may include a modem, a network card (wireless or wired), an infrared communication device, a wireless communication device (such as an antenna), and/or chipset (such as a Bluetooth® device, an 802.6 device (e.g., Metropolitan Area Network (MAN)), a WiFi device, a WiMax device, cellular communication facilities, etc.), and/or the like. The communication unit **612** may permit data to be exchanged with a network and/or any other devices or systems described in the present disclosure.

[0107] In some embodiments, the display **610** may be configured as one or more displays, like a liquid crystal display (LCD), light emitting diodes (LED), Braille terminal, or other type of display. In some embodiments, the display **610** may include a touch sensitive display. For example, the display **610** may include a display and/or a screen that is resistive (e.g., pressure sensitive), capacitive (e.g., touch sensitive), projected capacitive, or that uses acoustic waves and infrared signals.

[0108] The sensor **616** may include one or more devices and/or sensors for inside-out tracking of the computing system **600**. In some embodiments, the sensor **616** may be implemented using any suitable number (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, etc.) of degrees of freedom. In these and other embodiments, the different types of sensors may vary depending on a number of degrees of freedom. For example, in six degrees of freedom (6DoF) and/or six-axis applications, the sensor **616** may include accelerometers and gyroscopes, while in nine-axis applications, the sensor **616** may include accelerometers, gyroscopes, and magnetometers. In some embodiments, the sensor **616** may determine the orientation of the computing system **600**.

[0109] The input interface **608** may include one or more control mechanisms to allow the user to

send commands and/or requests to the computing system **600**. The commands and/or the requests may relate to certain actions. In some embodiments, the one or more control mechanisms may include any suitable input mechanisms. For example, the one or more mechanisms may include triggers, analog sticks, touchpads, buttons, squeeze grips, d-pads, etc. In some embodiments, the input interface **608** may include any other suitable types of input devices. For example, the input interface **608** may include a microphone configured to receive voice input.

[0110] In some embodiments, the input interface **608** may include an on-screen interface shown via the display **610**. For example, in instances in which the display **610** is touch sensitive, an on-screen interface may be displayed. In these and other embodiments, the user may interact with the computing system **600** via the on-screen interface.

[0111] In the present disclosure, the sensor **616** and/or the processor **602** may be implemented using hardware including one or more processors, central processing units (CPUs) graphics processing units (GPUs), data processing units (DPUs), parallel processing units (PPUs), microprocessors (e.g., to perform or control performance of one or more operations), field-programmable gate arrays (FPGA), application-specific integrated circuits (ASICs), accelerators (e.g., deep learning accelerators (DLAs), optical flow accelerators (OFAs)), programmable vision accelerators (including one or more direct memory address (DMA) systems and/or vector processing units (VPUs)), and/or other processor types. In some other instances, the one or more modules may be implemented using a combination of hardware and software. In the present disclosure, operations described as being performed by a respective module may include operations that the respective module may direct a corresponding computing system to perform.

[0112] Modifications, additions, or omissions may be made to FIG. **6** without departing from the scope of the present disclosure. For example, the computing system **600** may include more or fewer elements than those illustrated and described in the present disclosure.

[0113] The use of the terms “substantially” or “generally” means fundamentally, largely, or mostly and allow for deviations. For example, substantially circular means exactly circular, mostly circular, or largely circular in appearance. As another example, substantially centered means an object is exactly centered or close to centered in appearance.

[0114] In accordance with common practice, the various features illustrated in the drawings may not be drawn to scale. The illustrations presented in the present disclosure are not meant to be actual views of any particular apparatus (e.g., device, system, etc.) or method, but are merely idealized representations that are employed to describe various embodiments of the disclosure. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may be simplified for clarity. Thus, the drawings may not depict all of the components of a given apparatus (e.g., device) or all operations of a particular method.

[0115] Embodiments described in the present disclosure may be implemented using computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable media may be any available media that may be accessed by a general purpose or special purpose computer. By way of example, and not limitation, such computer-readable media may include non-transitory computer-readable storage media including Random Access Memory (RAM), Read-Only Memory (ROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), Compact Disc Read-Only Memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices, flash memory devices (e.g., solid state memory devices), or any other storage medium which may be used to carry or store desired program code in the form of computer-executable instructions or data structures and which may be accessed by a general purpose or special purpose computer. Combinations of the above may also be included within the scope of computer-readable media.

[0116] Computer-executable instructions may include, for example, instructions and data, which cause a general-purpose computer, special purpose computer, or special purpose processing device

(e.g., one or more processors) to perform a certain function or group of functions. Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are described as example forms of implementing the claims.

[0117] Terms used in the present disclosure and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including, but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes, but is not limited to,” etc.).

[0118] Additionally, if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations.

However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

[0119] In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” or “one or more of A, B, and C, etc.” is used, in general such a construction is intended to include A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together, etc.

[0120] Further, any disjunctive word or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

[0121] All examples and conditional language recited in the present disclosure are intended for pedagogical objects to aid the reader in understanding the present disclosure and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present disclosure have been described in detail, various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the present disclosure.

Claims

1. A system comprising: a sensor configured to generate sensor data based on a factor of the system; a display configured to display an animated face of an artificial intelligence (AI) agent; a processor coupled to the sensor and the display, the processor configured to: determine an orientation of the system based on the sensor data; determine an orientation of a face of a user relative to the system based on the sensor data; and control rotation of the animated face on the display based on the orientation of the system and the orientation of the face of the user to maintain a consistent orientation of the animated face on the display relative to the face of the user.
2. The system of claim 1, wherein the display is configured to display the animated face of the AI agent such that the animated face occupies an entirety of a viewable area of the display.
3. The system of claim 1 further comprising a housing configured to house the display, wherein: the housing defines a viewable area of the display; and the display is configured to display the AI agent

such that the animated face occupies an entirety of the viewable area such that the housing forms a portion of a head of the AI agent.

4. The system of claim 1, wherein: the animated face of the AI agent includes a facial feature; the system comprises an input device, the input device configured to capture user input and generate input data representative of the user input; the processor is configured to: receive the input data; identify an output of the system based on the input data; and animate the facial feature of the animated face based on the identified output of the system such that the facial feature is synchronized with the output of the system.

5. The system of claim 4, wherein: the facial feature comprises at least one of a mouth or a face; the system further comprises a memory to store viseme data representative of positions of the mouth or the face when speaking various words; and the processor is configured to: identify a word included in the output of the system; and animate the mouth or the face of the AI agent based on the viseme data such that the mouth or the face are synchronized with the word as part of the output of the system.

6. The system of claim 1, wherein: the system comprises a housing configured to house the display, at least a portion of the housing comprising a substantially circular shape; the display comprises a viewable area that includes a substantially circular shape; the display is configured to display the animated face of the AI agent within the viewable area such that the animated face is substantially centered around a point of the display; and the processor is configured to control the rotation of the animated face such that the animated face rotates around the point of the display to maintain substantially consistent distances between individual facial features of the animated face and a boundary of the viewable area or edges of the housing when the system is at different orientations.

7. The system of claim 1, wherein: the animated face of the AI agent includes a facial feature; the processor is configured to: identify an output of the system; determine an emotion corresponding to the output of the system; and animate the facial feature to show the emotion corresponding to the output of the system.

8. The system of claim 1, wherein the animated face comprises eyes; and the processor is configured to: track movement of eyes of the user based on the sensor data; and control a gaze of the eyes of the animated face based on the movement of the eyes of the user such that the animated face maintains eye contact with the eyes of the user.

9. The system of claim 1, further comprising a microphone configured to capture audio input and generate audio data representative of the audio input, wherein the processor is configured to: determine the audio input comprises a wake command based on the audio input; and cause the display to transition from a standby mode to an active mode and cause the animated face of the AI agent to be displayed.

10. The system of claim 1 further comprising a memory configured to store at least one of sensor data, viseme data, output data, or input data.

11. The system of claim 1, wherein the sensor comprises at least one of an accelerometer, an inertial measurement unit (IMU), a magnetometer, a gyroscope, a magnetic compass, a light detection and ranging sensor, or a camera.

12. The system of claim 1, wherein the processor controls the rotation of the animated face on the display to permit the user to position the system in different positions based on a preference of the user rather than factors of the display or the system.

13. A system comprising: a sensor configured to generate sensor data based on a factor of the system; a display configured to display an animated face of an artificial intelligence (AI) agent such that the animated face occupies an entirety of a viewable area of the display, the animated face comprising a facial feature; a housing configured to house the display, the housing defining the viewable area of the display; and a processor coupled to the sensor and the display, the processor configured to: determine an orientation of the system based on the sensor data; determine an orientation of a face of a user relative to the system based on the sensor data; position the animated

face on the display based on the orientation of the system and the orientation of the face of the user to maintain a consistent orientation of the animated face on the display relative to the face of the user; identify an output of the system; and animate the facial feature of the animated face based on the identified output of the system such that the facial feature is synchronized with the output of the system.

14. The system of claim 13, wherein: the facial feature comprises at least one of a mouth or a face; the system further comprises a memory configured to store viseme data representative of positions of the mouth or the face when speaking various words; and the processor is configured to: identify a word included in the output of the system; and animate the mouth or the face of the AI agent based on the viseme data such that the mouth or the face are synchronized with the word as part of the output of the system.

15. The system of claim 13, wherein the processor is configured to: determine an emotion corresponding to the output of the system; and animate the facial feature to show the emotion corresponding to the output of the system.

16. The system of claim 13, wherein the animated face comprises eyes; and the processor is configured to: track movement of eyes of the user based on the sensor data; and control a gaze of the eyes of the animated face based on the movement of the eyes of the user such that the animated face maintains eye contact with the eyes of the user.

17. The system of claim 13, further comprising a microphone configured to capture audio input and generate audio data representative of the audio input, wherein the processor is configured to: determine the audio input comprises a wake command based on the audio input; and cause the display to transition from a standby mode to an active mode and cause the animated face of the AI agent to be displayed.

18. The system of claim 13 further comprising a memory configured to store at least one of sensor data, viseme data, output data, or input data.

19. The system of claim 13, wherein the sensor comprises at least one of an accelerometer, an inertial measurement unit (IMU), a magnetometer, a gyroscope, a magnetic compass, a light detection and ranging sensor, or a camera.

20. The system of claim 13, wherein the processor controls the position of the animated face on the display to permit the user to position the system in different positions based on a preference of the user rather than factors of the display or the system.
